

04 Benchmark and Ablation Report

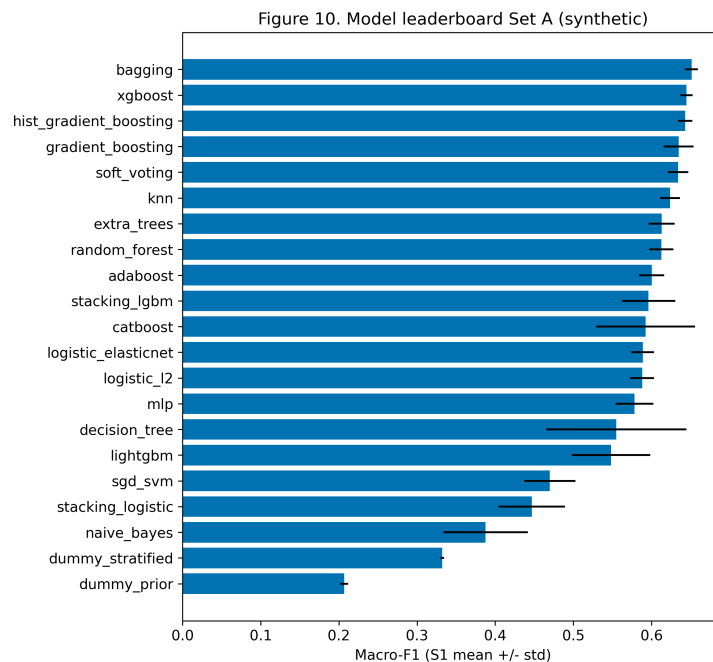
Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

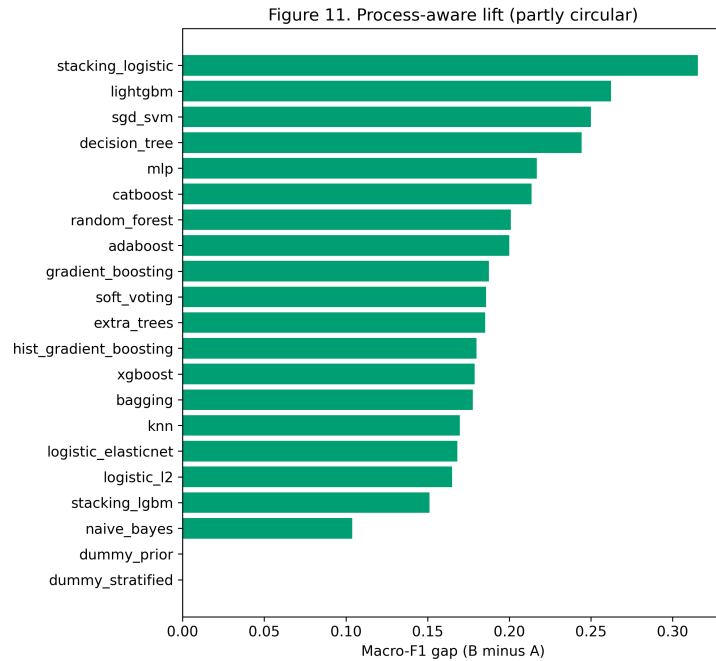
1. Split strategies

S1: StratifiedGroupKFold by district (primary). S2: random stratified (optimistic). S3: temporal train through 2020, test 2021 to 2024. S4: leave-one-state-out. Locked district-held-out test used once per model.

2. Champion

```
{
  "model": "bagging",
  "feature_set": "A",
  "s1_macro_f1_mean": 0.6512228207775592,
  "s1_macro_f1_std": 0.008133974052273845,
  "s1_log_loss_mean": 0.7122904938286985,
  "s3_macro_f1": 0.41142453278282903,
  "s3_accuracy": 0.4625675522467027,
  "final_macro_f1": 0.5776477399091471,
  "final_accuracy": 0.67804531986813,
  "final_log_loss": 0.6826631244589397,
  "latency_ms_per_1k": 10.341655886016342,
  "train_seconds": 1.643704299989622
}
```





3. Extended tests

McNemar (champion vs runner-up): {"a": "bagging", "b": "lightgbm", "statistic": 449.5316766211604, "pvalue": 9.12086070785514e-100, "n01": 3661, "n10": 5715}.

Bootstrap 95% CI macro-F1: {"bagging": {"mean": 0.5776749385805315, "ci_low": 0.5717586544924682, "ci_high": 0.5838244265750968}, "lightgbm": {"mean": 0.5921925012176992, "ci_low": 0.5864329420439558, "ci_high": 0.5981456560467499}, "n_boot": 1000}

Friedman/Nemenyi: {"friedman_stat": 27.400000000000006, "friedman_p": 0.0002823036951306332, "models": ["lightgbm", "xgboost", "random_forest", "hist_gradient_boosting", "catboost", "soft_voting", "logistic_l2", "bagging"], "nemenyi_pvalues": {"lightgbm": {"lightgbm": 1.0, "xgboost": 0.0597, "random_forest": 0.9695, "hist_gradient_boosting": 0.0274, "catboost": 0.8483, "soft_voting": 0.4375, "logistic_l2": 0.9999, "bagging": 0.0027}, "xgboost": {"lightgbm": 0.0597, "xgboost": 1.0, "random_forest": 0.5255, "hist_gradient_boosting": 1.0, "catboost": 0.7805, "soft_voting": 0.9858, "logistic_l2": 0.1624, "bagging": 0.9858}, "random_forest": {"lightgbm": 0.9695, "xgboost": 0.5255, "random_forest": 1.0, "hist_gradient_boosting": 0.3548, "catboost": 0.9999, "soft_voting": 0.9695, "logistic_l2": 0.9982, "bagging": 0.9982}, "hist_gradient_boosting": {"lightgbm": 0.0274, "xgboost": 1.0, "random_forest": 0.3548, "hist_gradient_boosting": 1.0, "catboost": 0.9999, "soft_voting": 0.9695, "logistic_l2": 0.9982, "bagging": 0.9982}, "catboost": {"lightgbm": 0.8483, "xgboost": 0.7805, "random_forest": 0.9999, "hist_gradient_boosting": 0.9999, "catboost": 1.0, "soft_voting": 0.9695, "logistic_l2": 0.9982, "bagging": 0.9982}, "soft_voting": {"lightgbm": 0.4375, "xgboost": 0.9858, "random_forest": 0.9695, "hist_gradient_boosting": 0.9695, "catboost": 0.9695, "soft_voting": 1.0, "logistic_l2": 0.9982, "bagging": 0.9982}, "logistic_l2": {"lightgbm": 0.9999, "xgboost": 0.1624, "random_forest": 0.9982, "hist_gradient_boosting": 0.9982, "catboost": 0.9982, "soft_voting": 0.9982, "logistic_l2": 1.0, "bagging": 0.9982}, "bagging": {"lightgbm": 0.0027, "xgboost": 0.9858, "random_forest": 0.9982, "hist_gradient_boosting": 0.9982, "catboost": 0.9982, "soft_voting": 0.9982, "logistic_l2": 0.9982, "bagging": 1.0}}

Figure 14. S1 vs S3 (synthetic)

